

Telnet

- Used to access the switches and routers remotely
- Clear text protocol
- Port 23
- Telnet ip

SSH

- Access devices remotely in a secured encrypted way
- Uses port 22
- SSH IP



Admin



```
(config)#username cloudecode passowrd xyz
(config)#line vty 0 15
(-line)#Transport input telnet/ssh
(-line)#login local
```

FTP

- File Transfer protocol
- Reliable on TCP
- Port 20 for data transfer
- Port 22 for control session

TFTP

- UDP port 69



Admin



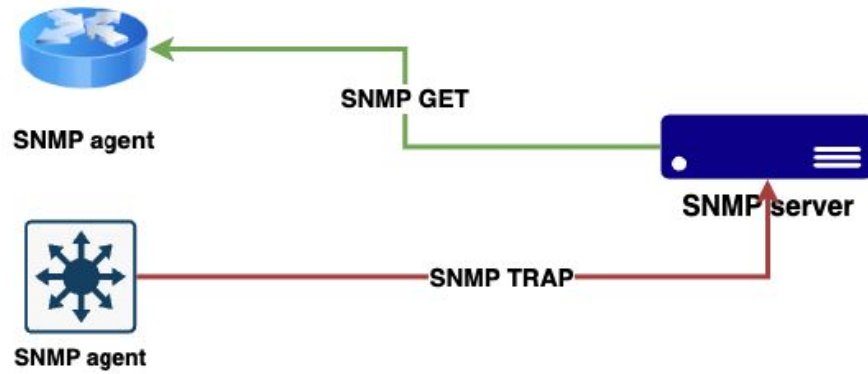
FTP server

Protocol Name	Port Number(s)	Description
HTTP	80	Used for transferring web pages over the internet.
HTTPS	443	Secure version of HTTP using encryption (SSL/TLS).
FTP	20 (Data), 21 (Control)	File Transfer Protocol for transferring files.
SSH	22	Secure Shell for encrypted remote login and file transfer.
Telnet	23	Remote login protocol (unencrypted).
DNS	53	Resolves domain names to IP addresses (and vice versa).
SMTP	25	Used to send emails (Simple Mail Transfer Protocol).
SNMP	161 (Queries), 162 (Traps)	Simple Network Management Protocol for network monitoring.
RDP	3389	Remote Desktop Protocol for accessing desktops remotely.
NTP	123	Network Time Protocol for clock synchronization.

SNMP

- **Simple Network Management Protocol** UDP (default port **162** for traps).
- **Monitor all network devices**
- **Get interface / CPU / Memory info, high CPU+ Configuration**
- **SNMP server query information from SNMP agent using Get**
- **or SNMP agent send Trap with info to the server**
- **Version 2: should be acked + uses community**
- **Version3: enhanced security (auth + encryption)**

• `snmp-server community <community-string> R0`



Severity Level	Name	Description
0	Emergency	System is unusable (e.g., kernel panic).
1	Alert	Action must be taken immediately (e.g., database down).
2	Critical	Critical conditions (e.g., hardware failure).
3	Error	Non-critical error messages.
4	Warning	Warning conditions (e.g., nearing disk space limit).
5	Notification	Normal but significant events (e.g., interface up/down).
6	Informational	Informational messages (e.g., routing table updates).
7	Debug	Debug-level messages (e.g., troubleshooting details).

Syslog

- **On Router logs are saved in buffer memory**
- **better to be sent to a server, Operates over UDP (port 514)**
- **# show logging**
- **if interface goes down, Configuration changes, error logs**
- **Emergency, Alert, Critical, Error, Warning, Notification, Info, Debug**



- `logging <syslog-server-ip-address>`
- `logging trap <level>`

NTP

- Network Time protocol
- over UDP (port 123)
- Uses external server to Sync time
- You can configure the router as NTP server



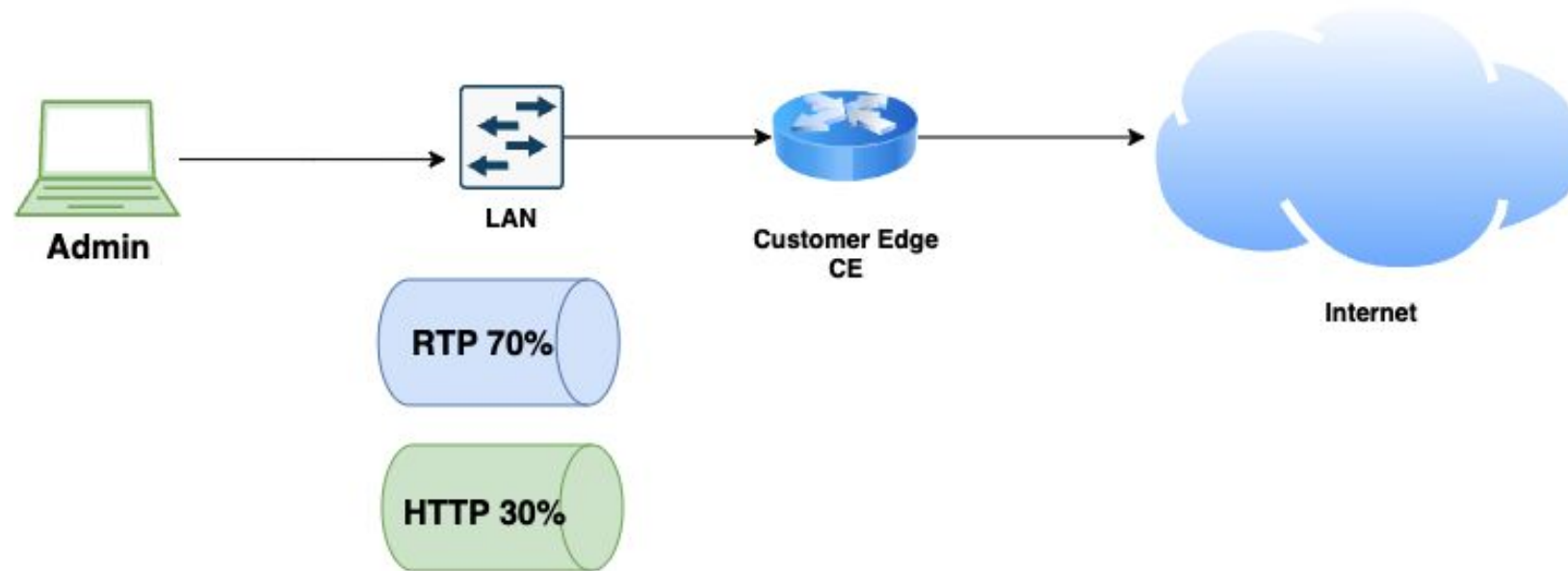
#clock set HH:MM:SS DAY MONTH YEAR

#ntp server ip version 4

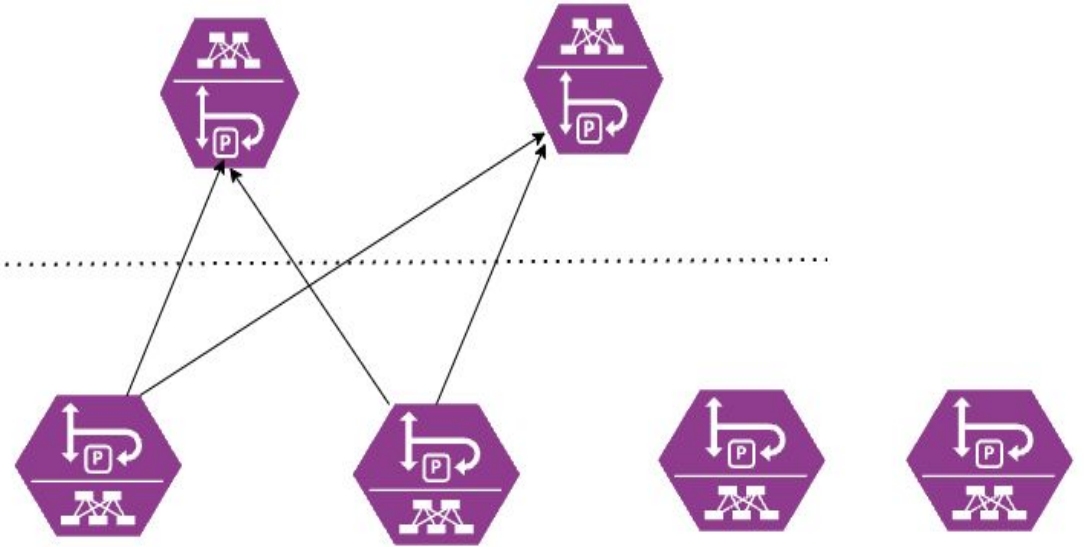
#ntp server 192.168.1.101 prefer

QOS

- **Quality of service**
- BW congestion happens when WAN BW is less than traffic comes from LAN
- **Classification: Classify Traffic, RTP/ HTTP**
- **Marking: Mark type of traffic with DSCP values or IP precedence**
- **Queuing: place every kind of traffic in corresponding Queue**
- **policing: limit BW in specific queue**



Design



Physical layer

- **copper:** used for ethernet, 4 pairs of copper
 - speed 100M, 1G, 10G
 - Connector: RJ45,
 - 1000 base T
-
- **Fiber:** uses light to send traffic
 - Multimode (short distance 500m)
 - SingleMode (long distance 80km)
 - Connector SFP
 - 1000 base S/L



1. White-Orange
2. Orange
3. White-Green
4. Blue
5. White-Blue
6. Green
7. White-Brown
8. Brown



SMF

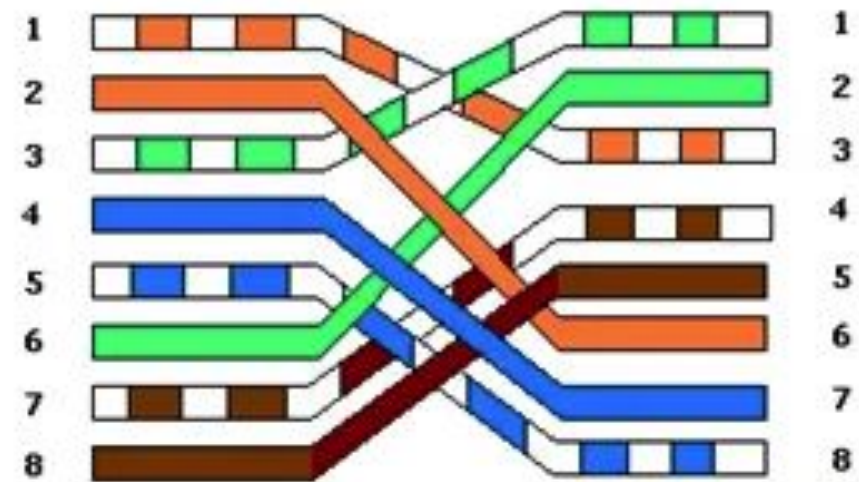
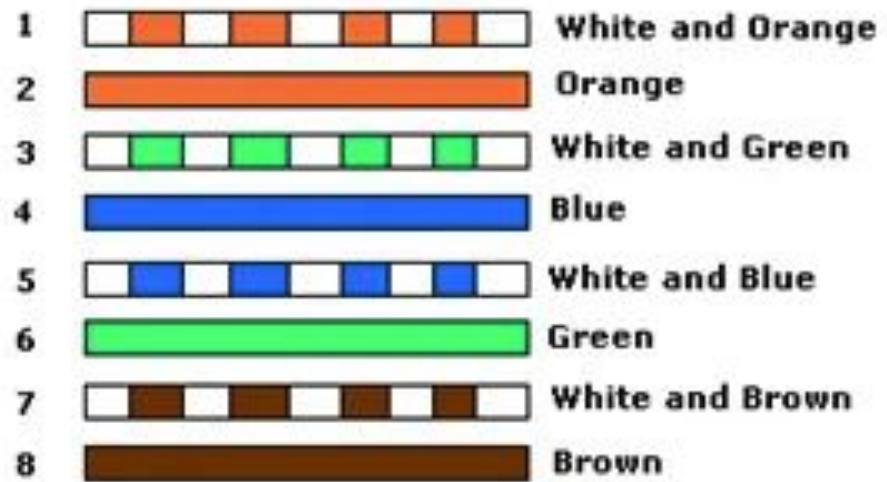


MMF



MMF

Source: <https://www.fs.com/>



Type	Use Case	Advantages	Limitations
Copper (Twisted Pair)	- Short-distance connections ($\leq 100\text{m}$) in LANs (e.g., office PCs, switches). - Low-cost networks.	- Affordable and easy to install. - Supports Power over Ethernet (PoE).	- Limited to 100 meters. - Susceptible to interference.
Multimode Fiber (MMF)	- Medium-distance connections in data centers ($\leq 500\text{m}$). - High-speed LANs or storage networks.	- Cheaper than SMF. - Easy to deploy for short distances.	- Limited distance due to modal dispersion.
Single-Mode Fiber (SMF)	- Long-distance connections (up to 80km or more) in WANs. - High-bandwidth backbones and ISPs.	- Ideal for long distances and high bandwidth. - Immune to interference.	- Expensive cables and transceivers. - Harder to terminate.

IPv6

- Hexa Decimal, 8 Fields each field is 4 Hexa
- 128 bit -----
- Network : Node / mask
- Leading Zero are removed ----:0003:-----
- A213:B123:0000:0000:C:10:5:3ABA == A213:B123::C:10:5:3ABA

Types

Global Unicast

2000:: /3

Public on Internet

Unique Local

FC00::/7

Private range on Internet

Link Local

FE80::0/10

Not routable anywhere

Multicast

FF00::0/8

Multicast

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3

Decimal	Binary	Hexadecimal
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Configuration

R1#configure terminal

R1(config)#ipv6 unicast-routing

R1(-int)#interface GigabitEthernet0/0

R1(-int)#ipv6 address 2001:db8:1::1/64

R1(-int)#ipv6 ospf 1 area 0

R1(config)#ipv6 route ::/0 2001:db8:2::2

2001:db8:2::2



R1



R2



Internet

2001:db8:abcd:0001::/64

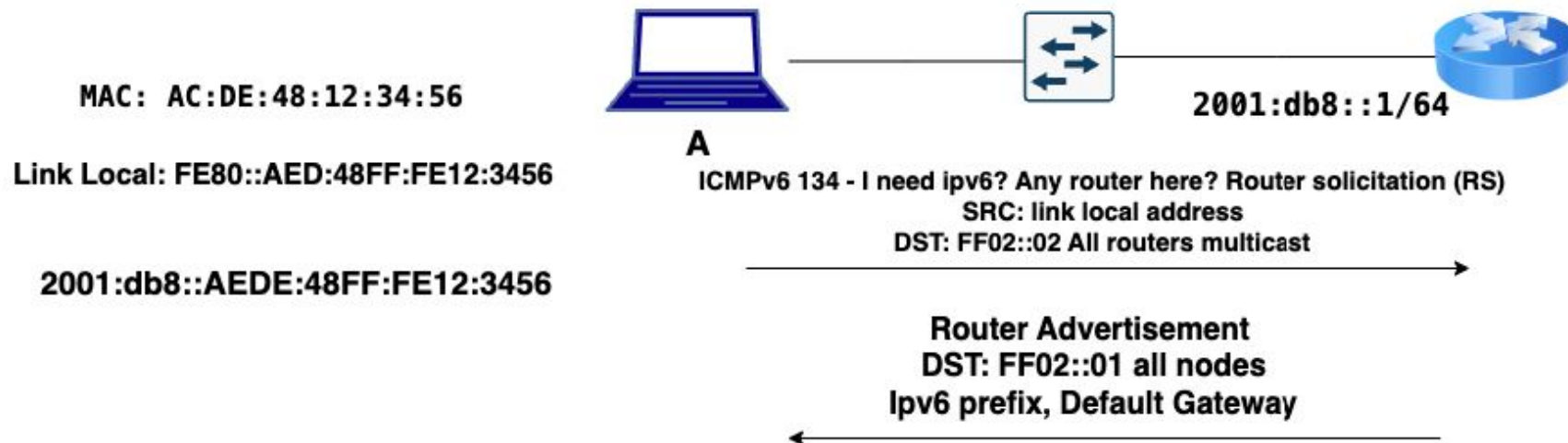


2001:db8:abcd:0001::

2001 -> 0010 0000 0000 0001
0db8 -> 0000 1101 1011 1000
abcd -> 1010 1011 1100 1101
0001 -> 0000 0000 0000 0001

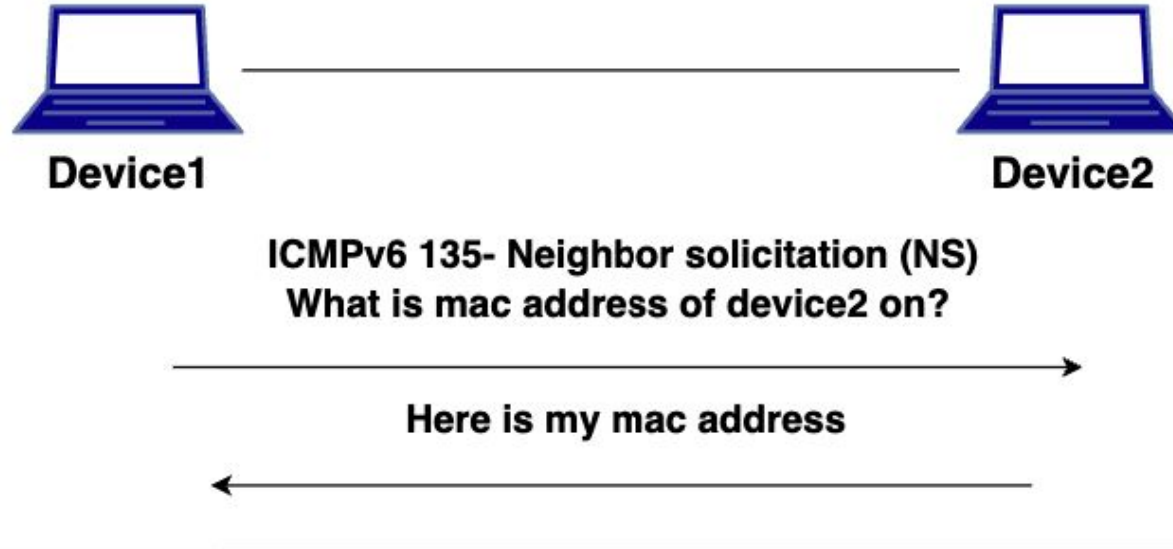
Stateless Address auto configuration - SLAAC

- It generates a **link-local address** automatically (using the FE80::/10 prefix and its MAC address in EUI-64 format).
- **Host Sends a Router Solicitation (RS) Message**
- Routers periodically send Router Advertisement (RA) messages or respond directly to RS messages.
- RA message contains important information, such as:
 - **Prefixes:** The router advertises the IPv6 prefix (e.g., 2001:db8::/64) for the network + gateway
- The **Interface Identifier** can be derived in one of two ways:
 - **EUI-64 Format:** Based on the node's MAC address.
 - The MAC address (48 bits) is expanded into 64 bits by adding the **FF:FE** segment in the middle. AC:DE:48:12:34:56



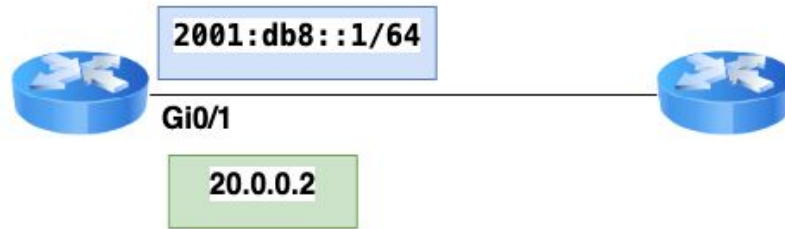
Neighbor Discovery Protocol

- It replaces ARP (Address Resolution Protocol) used in IPv4
- Discovering other devices on the same network & get MAC address.
- Neighbor Solicitation (NS):
 - Sent by a host to resolve the MAC address of another host
- Neighbor Advertisement (NA):
 - Contains the MAC address of the responding host



Dual Stack

- a network interface or device is configured to use both **IPv4** and **IPv6**



6in4 Tunneling

- Configuring a tunnel between two IPv6-enabled routers over an IPv4-only network.



NAT64

- Allows IPv6-only devices to communicate with IPv4-only devices by translating IPv6 addresses to IPv4 and vice versa.

